```
    delete b.lb1;
    return 0;
}
```

Scenario 2:

```
#include <iostream>
#include <string>
#include <vector>

using namespace std;

class calendar {
private:
    int year;
    vector<vector<string>> tasks;

    bool isLeapYear() const {
```

Dated:

```
return ( year % 4 == 0 && year % 100 != 0) ||
                (year % 400 == 0 ); }

public :
  Calendar (int y ) {
     year = y;
     tasks . resize (12);
    for (int i =0; i <12; ++i ) {
          int days = getDaysInMonth( i+1 );
    tasks [i]. resize (days, " "); }
              }

int getDaysInMonth ( int month) const {
    switch (month) {
    case 1: return 31;
    case 2: return isLeapYear () ? 29 : 28;
   case 3 : return 31;
   case 4 : return 30;
   case 5 : return 31;
   case 6 : return 30;
   case 7 : return 31;
   case 8 : return 31;
   case 9: return 30;
   case 10 : return 31;
   case 11 : return 30;
   case 12 : return 31;
   default : return 0;
     }
```

Dated:

```
bool isValidDate(int month, int day)
const {
    if (month < 1 || month > 12)
return false;
    int maxDay =
getDayInMonth(month);
    if(day < 1 || day > maxDay)
return false;
    return true;
}
bool hasTask(int month, int Day) const
{
    if(!isValidDate(month, day))
return false;
    return !tasks[month-1][day-1].empty();
}
void addTask(int month, int day, string description)
{
if(!isValidDate(month, day)){
    cout << "Error: Invalid Date entered ("<< month
        << "/"<< day <<").
Task rejected." << endl;
    return;
}
if(hasTask(month, day)){
    cout << "Error: Date "<< month << "/"<< day<<
        "already contains a task. Can't overri
                                    << endl;
    return; }
```

Dated:

```
        tasks[month-1][day-1] =
description;
        cout << "Success : Task added for " << month <<
            "/" << day << "." << endl;
    }
void removeTask(int month, int day) {
    if (!isValidDate(month, day)) {
        cout << "Error : Invalid date entered." << endl;
        return; }
if (!hasTask(month, day)) {
    cout << "Message : No task exists on " << month <<
    "/" << day << " to remove" << endl;
        return; }
tasks[month-1][day-1] = "";
    cout << "Success : Task removed from " << month <<
            "/" << day << "." << endl; }
void displayAllTasks() const {
    cout << "In scheduled tasks for year " << year <<
            "\n ====\n";
bool foundAny = false;
for (size_t i=0; i<tasks.size(); ++i) {
    for (size_t j=0; j<tasks[i].size(); ++j) {
        if (!tasks[i][j].empty()) {
            cout << "Month " << i+1 << ", Day " << j+1 <<
                    ": " << tasks[i][j] << endl;
            foundAny = true;
        }
    }
}
if (!foundAny) {
```

Dated:

```
        cout << " No tasks scheduled in the calendar. " << endl;
    }

int main() {
    int year;
    cout << " Enter the year for the calendar : ";
    cin >> year;
    Calendar myCalendar(year);
    int choice;

    do {
cout << "\n Digital Task Management Menu " << endl;
cout << " 1. Add a task " << endl;
cout << " 2 Remove a task " << endl;
cout << " 3 Display all tasks " << endl;
cout << " 4 Exit " << endl;
cout << " Enter your choice: " ;
cin >> choice;

if (choice == 1) {
    int m, d;
    string desc;
    cout << "Enter month (1-12): ";
    cin >> m;
    cout << " Enter day : " ;
    cin >> d;
    cin.ignore();
cout << "Enter task description : ";
getline(cin, desc);
myCalendar.addTask(m, d, desc); }
```

Dated:

```
else if (choice == 2) {
    int m, d;
    cout << "Enter month (1-12): ";
    cin >> m;
    cout << "Enter day: ";
    cin >> d;
    myCalendar.removeTask(m, d);
} else if (choice == 3) {
    myCalendar.displayAllTasks();
} else if (choice == 4) {
    cout << "Exiting system. Goodbye!" << endl;
} else {
    cout << "Invalid choice!" << endl;
}
} while (choice != 4);

return 0;
}
```